

OpenVLA-OFT · arXiv 2502.19645 · Kim, Finn, Liang (Stanford)

Archaeologist

11+3

OFT || FAST || FM forks · ACT chunk+L1 ancestry
Do not collapse names

Oct 26 paper club · VLA Architectures

SPINE: AR discrete bins → parallel continuous L1 chunks

OFT ≠ FAST ≠ FM · FT recipe / speed+success patch · not $\pi_{0.5}$ on BEHAVIOR

TAXONOMY · $\text{OFT} \neq \text{FAST} \neq \text{FM}$

Three parallel improvement paths — do not collapse names

OpenVLA-OFT

FT recipe

Parallel continuous
L1 chunks
K=8 / 25

Decode: 1-pass
Loss: L1
Still 7B OpenVLA

FAST

Tokenizer

DCT→BPE discrete
chunk tokens
still AR

Decode: serial AR
Loss: CE
~750 ms/chunk class

π_0 / $\pi_{0.5}$

FM deploy

Continuous FM
chunks / NFE
OpenPI stack

Decode: denoise
Loss: FM / ϵ
 π_0 ALOHA ~291 Hz

Vanilla OpenVLA = AR discrete bins @ ~4.2 Hz → OFT rewrites output signature

OFT does NOT replace $\pi_{0.5}$ for 2026 BEHAVIOR · $\text{OFT} \neq \text{FAST}$ tokenizer swap

FORK GENEALOGY · OFT || FAST || FM

Archaeologist — parallel improvement paths; do not collapse



SIGNATURE SHIFT DIAGRAM

AR discrete single-step → parallel continuous L1 chunks

VANILLA OpenVLA

img + lang → VLM (causal)

AR next-token × D(=7)

7 × {0..255} bins → dequantize

single EE delta · K=1

~4.2 Hz LIBERO · ~1.8 Hz ALOHA

OFT

OpenVLA-OFT

img[+wrist][+proprio] + lang

bidirectional action queries

one forward → $a \in \mathbb{R}^{K \times D}$

continuous L1 MLP · K=8/25

~109.7 Hz · ALOHA ~77.9 Hz

FAST (sibling)

compress chunk → fewer
AR tokens; still sequential
~750 ms chunk class

π_0 / DP / ACT

continuous chunk experts
FM / diffusion / CVAE+L1
smaller bodies → higher Hz

OFT+

OFT + FiLM on both
vision towers
ALOHA multi-view lang

EXACT OFT RECIPE

Sec V-E / App. A — six ingredients (OFT+ = #6)

1

Parallel decoding

Bidirectional attn over
empty action embeds

2

Action chunking

K=8 LIBERO /
K=25 ALOHA

3

Continuous MLP head

4-layer ReLU;
drop 256-bin LM head

4

L1 on [-1,1]

Mean L1 regression
(train until L1 < 0.01)

5

LoRA r=32

Optional wrist +
proprio projector

6

OFT+: FiLM

Lang $\rightarrow \gamma, \beta$ on SigLIP
+ DINOv2 blocks

OUTPUT SIGNATURE SHIFT

Vanilla OpenVLA → OpenVLA-OFT

Axis	Vanilla OpenVLA	OFT
Decode	Serial AR discrete bins	Parallel continuous decode
Chunk	K=1	K=8 LIBERO / K=25 ALOHA
Head	256-bin CE	Continuous MLP + L1
Hz	~4.2	~109.7 LIBERO (26×); ALOHA ~77.9 (43×)
LIBERO SR	76.5%	97.1% (π ₀ 94.2%)

OFT+ adds FiLM on SigLIP+DINOv2 (multi-view language) — without FiLM, language → ~chance

Fig.2 · AR vs parallel · discrete CE vs continuous L1/diffusion

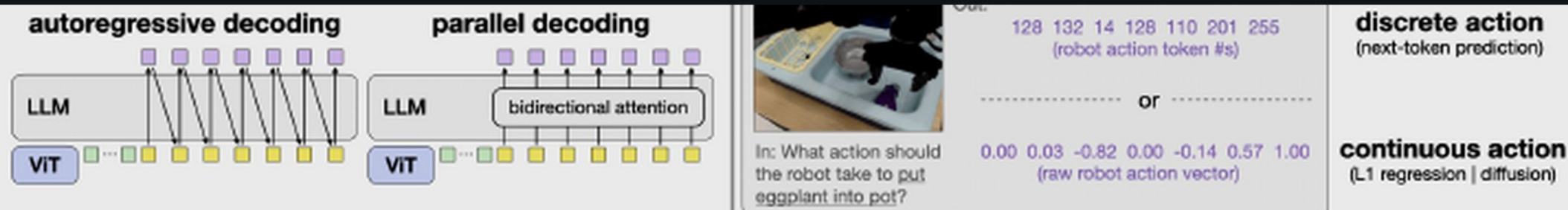


Fig. 2: Key design decisions for VLA fine-tuning. Left: Comparison between autoregressive decoding, which generates actions sequentially, and parallel decoding, which leverages bidirectional attention and generates all actions in a single forward pass. Right: Comparison between discrete action tokens with next-token prediction and continuous action values with L1 regression or diffusion modeling objectives. The original OpenVLA training scheme includes autoregressive decoding, discrete actions, and next-token prediction.

ing. While these diffusion-based VLAs achieve higher action throughput than autoregressive VLAs by generating multi-timestep action chunks simultaneously, they introduce computational trade-offs through slower training and multiple denoising or integration steps at inference time. Furthermore, these diffusion VLAs vary considerably in architecture, learning algorithm, vision-language fusion approach, and input-output specifications—and which design elements most significantly impact performance remains unclear. Through controlled experiments, we show that policies fine-tuned with a simpler L1 regression objective can match more complex approaches in

timestep action takes 0.33 seconds on an NVIDIA A100 GPU. For a chunk size of K timesteps and action dimensionality D , OpenVLA requires KD sequential decoder forward passes versus just D passes without chunking. This K -fold increase in latency makes action chunking impractical for high-frequency robots under the original formulation. In the next section, we present a parallel generation scheme that enables efficient action chunking.

IV. STUDYING KEY VLA FINE-TUNING DESIGN DECISIONS

Fig.1 · OFT+ ALOHA system · parallel chunk + FiLM

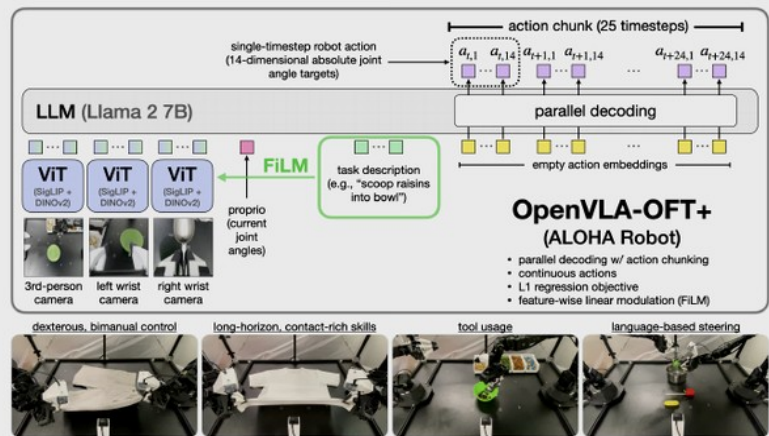


Fig. 1: OpenVLA-OFT+ on the bimanual ALOHA robot. Our Optimized Fine-Tuning (OFT) recipe enhances fine-tuned OpenVLA policies through improved inference efficiency, model quality, and input-output flexibility. The resulting OpenVLA-OFT+ policies execute diverse dexterous manipulation tasks on a real-world bimanual robot at high control frequencies (25 Hz). The “+” suffix indicates the integration of feature-wise linear modulation (FiLM) [37], which strengthens language grounding in tasks where accurate language understanding is critical for success.

denoting the augmented recipe as **OFT+**. OpenVLA-OFT+ successfully executes dexterous bimanual tasks like folding clothes and manipulating target food items based on the user’s prompt (see Figure 1), outperforming both fine-tuned VLAs (π_0 and RDT-1B) and prominent imitation learning policies trained from scratch (Diffusion Policy and ACT) by up to 15% (absolute) in average success rate. With 25-timestep action chunks, OpenVLA-OFT+ achieves $43\times$ faster throughput than base OpenVLA, demonstrating that our new fine-tuning recipe enables real-time robot control with strong task performance and language following ability.

II. RELATED WORK

Prior works have leveraged language and vision foundation models to enhance robotic capabilities, using them as pretrained visual representations that accelerate robotic policy learning [30, 33, 31, 20, 32], for object localization in robotics tasks [9, 47], and for high-level planning and reasoning [1, 17, 44, 16, 45, 18, 6]. More recently, researchers have explored fine-tuning vision-language models (VLMs) to directly predict low-level robotic control actions, producing “vision-language-action” models (VLAs) [4, 34, 24, 23, 7, 15, 8, 53, 58, 54, 3, 2], which have demonstrated effective language following [59] and generalization to out-of-distribution test conditions and unseen semantic concepts. These works focus primarily on model development, while we focus on developing a recipe for fine-tuning such models, justifying individual design decisions with insights that we gain from our empirical analysis.

Despite the importance of fine-tuning for real-world VLA deployment, empirical analysis of effective fine-tuning recipes remains limited. While Kim et al. [23] study various parameter update strategies and from their findings show that LoRA fine-tuning enables effective adaptation to single-arm robots operating at low control frequencies (< 10 Hz), their analysis does not extend to bimanual robots with high control frequencies (25-50+ Hz), a more complex control scenario. We address this gap by exploring VLA adaptation design decisions for fast inference and reliable task execution on a real-world bimanual manipulator with a 25 Hz controller.

Recent works by Belkhale and Sadigh [2] and Pertsch et al. [38] improve VLA efficiency through new action tokenization schemes, using vector quantization or discrete cosine transform-based compression to represent action chunks (sequences of actions) with fewer tokens than simple per-dimension binning (as used in RT-2 [4] and OpenVLA [23]). While these approaches achieve 2 to $13\times$ speedups for autoregressive VLAs, we explore design decisions beyond autoregressive modeling, which remains inherently limited by iterative generation. Our parallel decoding approach, when paired with action chunking, achieves significantly greater speedups: $26\times$ to $43\times$ throughput with much lower latency (0.07 ms for single-arm tasks with one input image and 0.321 ms for bimanual tasks with three input images).

Another line of research [54, 27, 3] demonstrates effective VLA fine-tuning for high-frequency, bimanual manipulation using generative approaches like diffusion or flow match-

Fig.2 crop · AR vs parallel decoding

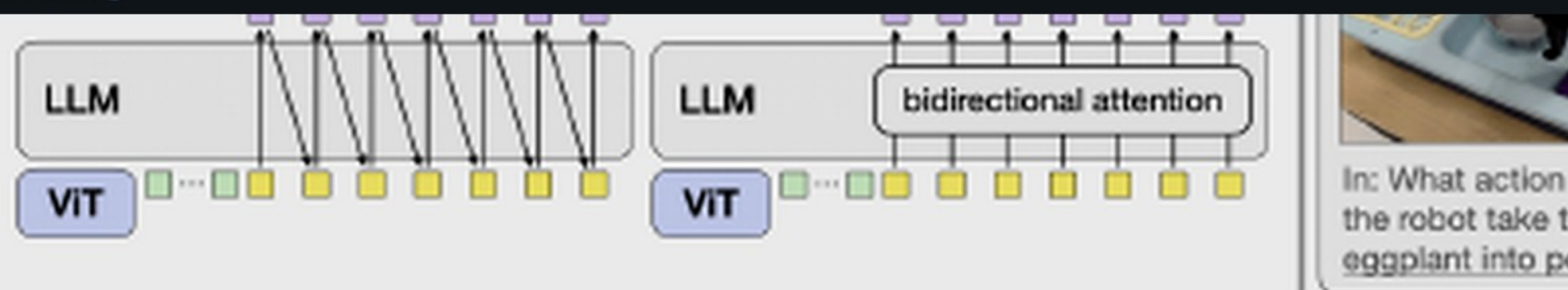


Fig. 2: Key design decisions for VLA fine-tuning. Left: Comparison between autoregressive decoding, which leverages bidirectional attention and generates all actions in a single forward pass, and next-token prediction and continuous action values with L1 regression or diffusion models. Right: Autoregressive decoding, discrete actions, and next-token prediction.

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CITE OFT FOR / DO NOT CITE FOR

Cite for

- ✓ FT recipe: parallel + chunk + Cont + L1
- ✓ Output signature shift vs vanilla
- ✓ High-Hz FT of large VLMs w/o LL controller
- ✓ Fair ablation morals vs Cont-Diff / PD&AC
- ✓ L1 $\approx$ diffusion SR at lower cost

Do not cite for

- ✗ Universal replacement for FM heads
- ✗ BEHAVIOR q-score SOTA
- ✗ Proof L1 models multimodal demos
- ✗ Fair ALOHA win without n/rubric/JAX notes
- ✗ Drop-in household champion

BOARD TABLE · Hz / SR

Memorize these cells — Stakeholder tattoo

LIBERO

SR avg	76.5% → 97.1%	+20.6 pp
Hz (A100)	4.2 → 109.7	26×
vs π_0 SR	π_0 94.2%	OFT wins
+wrist+proprio Hz	71.4	still $\gg$ vanilla

ALOHA (OFT+)

Hz	1.8 → 77.9	43×
π_0 Hz still	291.6	JAX faster
Scoop	100 vs π_0 93.3	OFT+ best
Put-in-pot	51 vs π_0 42	~half; hard

Latency (tables)

LIBERO Cont-L1 ~ 0.073 s
ALOHA OFT+ 0.321 s
(π_0 0.086 s)

Abstract caveat

“0.07 ms / 0.321 ms”
conflicts with Table II/III
Prefer seconds

Trainable

LIBERO OFT 279M
ALOHA OFT+ 853M
(FiLM +456M)

Rule: OpenVLA/HF/LoRA ops + high Hz → OFT mandatory; stock AR obsolete for 25-50 Hz

Max wall-clock + OpenPI/JAX → $\pi_0/\pi_{0.5}$ still win latency + pretrain coverage

Archaeologist · close

Parallel forks: OFT || FAST || FM. ACT supplies chunk+L1 ancestry.

Do not collapse names · OFT is FT recipe, not foundation model

Q&A · Oct 26 · OpenVLA-OFT / 2502.19645

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